

Conservation of Mass

Write the following investigation up using the standard scientific report scaffold.

INVESTIGATION 7.2

Conserve that mass!

AIM To compare the mass of the products of a chemical reaction with the mass of its reactants

Materials:

safety glasses
250 mL conical flask
4 Alka-Seltzer tablets
1 balloon
Matches
an electronic balance
100 mL measuring cylinder
water

METHOD AND RESULTS

CAUTION Wear safety glasses.

- ▶ Place the conical flask on the balance and pour in 100 mL of water.
- 1** Place two tablets alongside the conical flask and record the total mass.
- ▶ Remove the flask from the balance and drop the tablets into the water.
- 2** When the reaction is complete, weigh the flask and record the mass.
- ▶ Rinse out the flask thoroughly and again add 100 mL of water.
- ▶ Place two tablets inside the balloon. You may need to break the tablets into pieces to do this.
- ▶ Stretch the neck of the balloon over the conical flask, being careful not to drop the tablets into the water. The balloon should be flopped over, resting against the side of the flask.
- 3** Place the conical flask and balloon onto the balance and record the total mass.
- ▶ Lift up the top of the balloon and drop the tablets into the water in the conical flask.
- 4** When the reaction is complete, weigh the flask and record the mass. **Do not remove the balloon.**
- 5** After you have recorded the mass, remove the balloon. Light a match and test the gas in the conical flask. Record your observations.
- 6** Describe what happened during the reaction.

DISCUSS AND EXPLAIN

- 7** Which gas do you think filled the balloon?
- 8** Comment on your results of the total mass before and after each reaction. Explain your answer.
- 9** Why do you think it took a long time for the Law of Conservation of Mass to be developed?